

# Chapter 06

## IoA Networks

## Network Enumeration Activity

Lists systems on a network  
Identifies unauthorized devices  
Detects unknown users  
Supports network visibility

## Unauthorized Devices

New systems  
require investigation

Must be properly procured

Unapproved devices  
are suspicious

Indicates potential intrusion

## Anomalous Network Activity

Unusual behavior should  
be reviewed

Deviates from normal operations

May not always be malicious

Requires investigation

## Baseline Monitoring

Establish normal  
network behavior

Detect deviations from baseline

Supports anomaly detection

Improves monitoring accuracy

## Common Network Ports

Many ports available

Key ports to recognize

- 80 – HTTP
- 443 – HTTPS
- 22 – SSH

Used in network communication

## Indicators of Network Attacks

Multiple signs of activity

Requires monitoring and analysis

Detection based on anomalies

Key for threat identification

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IoA - IAM



# IAM Account Compromise

User accounts are  
primary targets

Attackers seek credential access

Enables unauthorized  
system entry

Leads to broader compromise

## Unauthorized Access

Access beyond normal privileges

Unexpected file or service usage

Deviates from user behavior

Indicates misuse of account

# Privilege Escalation

Elevates access beyond user level

Common attacker objective

Expands control within system

Enables further compromise

## Impossible Travel

Logins from distant locations

Occur within  
unrealistic timeframes

Indicates account compromise

Requires investigation

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## IoA Hosts

## Resource Consumption

Not all resource-heavy processes are malicious — browsers, video editors, and ML models are common culprits

- Investigate when usage starts interfering with normal operations

## Baseline Images and Unauthorized Configuration

Standard system images define the baseline for normal across an organization

Deviations like disabled security tools or non-company DNS signal unauthorized changes

## Unauthorized Software and Shadow IT

Unapproved installs raise red flags  
— unknown source,  
unknown vendor

Shadow IT: users bypassing policy  
without understanding the  
security risks



## Rogue Processes

Standardized OS images mean consistent processes — outliers are red flags

Attackers mimic names like svchost, swapping characters (O for zero) to blend in

## LOLBins

Attackers use built-in tools like PowerShell and cmd already on the system

- Legitimate admin tools won't trigger alerts, making them ideal for attackers

## Monitoring Critical Files

Monitor the Windows registry and hosts file for unauthorized changes

- The hosts file overrides DNS — attackers alter it to redirect traffic to malicious sites
- Entries resolve locally, bypassing authoritative DNS entirely

# Data Exfiltration

Attackers stage data in temp folders or the recycle bin before exfiltrating

- Sensitive files piled in one spot on a client warrants immediate investigation

## DNS Data Exfiltration

Stolen data is encoded as subdomains in DNS queries — e.g., secret.attacker.com

- Attackers review their DNS logs to extract the queried subdomains

# Chapter 06

## IoA- Cloud and Applications

# Service Disruption

**Cloud-hosted applications and services are increasingly targeted**

- Not every outage is malicious, but every outage should be investigated
- Maintaining availability requires proactive monitoring of all services

# Anomalous Activity

**Any out-of-the-norm behavior on cloud services deserves a closer look**

- It may not be malicious, but it's worth exploring for suspicious patterns
- Monitoring helps spot unusual traffic before it becomes a larger threat



## Resource Compromise

**Attackers can gain unauthorized access to your cloud environment**

- This applies to both local network and cloud-based assets equally
- Always be on the lookout for indicators of attack across all environments

# Chapter 06

## IoA – Email and Social Engineering

# Typosquatting

Attacker registers similar domain

Mimics legitimate websites

Exploits user input errors

Captures user credentials

## Typosquatting Attack Example

Slight domain misspellings

Users enter  
credentials unknowingly

Appears as legitimate service

Used to steal account access

## URL Shorteners

Hide destination addresses

Mask malicious links

Common in phishing attempts

Restricted in many organizations

## URL Lengtheners and Defense

Reveal actual destination URL

Increase user awareness

Help validate links

Reduce click risk

## Business Email Compromise BEC

Targeted email attack

Impersonates trusted contacts

Exploits business relationships

Designed to deceive users